

SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
DEPARTMENT OF ARTIFICIAL INTELLIGENCE.

B. Tech-3rd Year (5th Sem)

Data Science (AI303)

End-Sem Exam

Date: 25/11/2025

Time: 09:15 AM To 12:15 PM

Marks 100

Instructions: 1. Write your admission number/roll no. and other details clearly on the answer books and questions paper. 2. Support your answer with diagrams & examples wherever required.

Q. 1	An e-commerce company stores its transactional data in Hive. Two tables are provided: 1. Orders: orders(order_id INT, user_id INT, product_id INT, order_date STRING, amount DOUBLE) 2. Product: product(product_id INT, category STRING, price DOUBLE) Using the above schema, write HiveQL queries to answer the following: (a) Calculate the total sales amount for each product category by combining data from both tables. (b) Find the top 3 best-selling products based on the number of orders. (c) List all users who have purchased products from more than five different categories	[10] [CO2]
Q. 2	A movie review portal maintains two datasets in HDFS: 1. movies: movies(movie_id INT, title CHARARRAY, genre CHARARRAY) 2. ratings: ratings(user_id INT, movie_id INT, rating FLOAT) Using the above files, write Pig Latin scripts to perform the following tasks: (a) Join the movies and ratings datasets to display title, genre, and the corresponding avg_rating for each movie. (b) Determine the top 5 highest-rated movies based on their average rating. (c) Determine the top 5 highest-rated movies based on their average rating.	[10] [CO2]
Q. 3	Draw the architecture of Apache Hive and Pig and Explain its steps.	[10] [CO2]
Q.4 (a)	Write a short note on Locality Sensitive Hashing.	[5] [CO3]
Q.4 (b)	Compute the minhash signature for each column and calculate the similarity of signature of each column to other columns using Jaccard similarity; i.e. similarity between C1-C3, C2-C4, C1-C2, C3-C4 You need to consider 3 permutations of rows of a matrix as follows: P1: 3 4 7 6 1 2 5 P2: 4 2 1 3 6 7 5 P3: 1 3 7 6 2 5 4	[10] [CO4]

	<table><tr><th>Rows</th><th>C₁</th><th>C₂</th><th>C₃</th><th>C₄</th></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>2</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>3</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>4</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>5</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>6</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>7</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	Rows	C ₁	C ₂	C ₃	C ₄	1	1	0	1	0	2	1	0	0	1	3	0	1	0	1	4	0	1	0	1	5	0	1	0	1	6	1	0	1	0	7	1	0	1	0	
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Q.4 (c)	Calculate the number of distinct elements in the following datastream using Flajolet-Martin algorithm: 2, 3, 1, 1, 2, 5, 2, 1 and given a hash function $h(x) = (x + 2) \bmod 64$	[5] [CO3]																																								
Q. 5	a) The following table contains four market-basket transactions: <table><tr><th>TID</th><th>Items Purchased</th></tr><tr><td>T1</td><td>{f, a, c, d, g, m, p}</td></tr><tr><td>T2</td><td>{a, b, c, f, m, l, o}</td></tr><tr><td>T3</td><td>{b, f, h, o}</td></tr><tr><td>T4</td><td>{b, k, c, p}</td></tr></table> Assume a minimum support threshold of 3. Using the FP-Growth algorithm, answer the following: i) Compute the support count of all items and identify the frequent items (support ≥ 3). Arrange the frequent items in descending order of support. ii) Using the ordering obtained in part (a), construct the FP-Tree. b) Consider the frequent itemset mining problem based on the following transactions table of the supermarket. Find out items that has minimum support of 2. Also find out the association rules that has minimum confidence of 50 %. Solve the problem using a-priori algorithm that is based on candidate generation. <table><tr><th>Transactions</th><th>Items</th></tr><tr><td>1</td><td>Milk, Bread, Butter</td></tr><tr><td>2</td><td>Bread, Cheese</td></tr><tr><td>3</td><td>Bread, Jam</td></tr><tr><td>4</td><td>Milk, Bread, Cheese</td></tr><tr><td>5</td><td>Milk, Jam</td></tr><tr><td>6</td><td>Bread, Jam</td></tr><tr><td>7</td><td>Milk, Jam</td></tr><tr><td>8</td><td>Milk, Bread, Jam, Butter</td></tr><tr><td>9</td><td>Milk, Bread, Jam</td></tr></table>	TID	Items Purchased	T1	{f, a, c, d, g, m, p}	T2	{a, b, c, f, m, l, o}	T3	{b, f, h, o}	T4	{b, k, c, p}	Transactions	Items	1	Milk, Bread, Butter	2	Bread, Cheese	3	Bread, Jam	4	Milk, Bread, Cheese	5	Milk, Jam	6	Bread, Jam	7	Milk, Jam	8	Milk, Bread, Jam, Butter	9	Milk, Bread, Jam	[20] [CO4]										
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Q.6	You are developing a software for an online supermarket, and your goal is to recommend new items to users, but you don't want to recommend items that they already bought. Each item has a unique long integer id, but	[10] [CO5]																																								

	the supermarket has so many items (2^{64} items) that you cannot store in memory. Which of the method/algorithm is best suited for testing if the recommended item is unseen by the user? Also explain why?																															
Q.7	a) Explain user-based collaborative filtering. (b) Using the following ratings matrix, compute the similarity scores and predict the missing rating for User A on Item I3 using cosine similarity. Show all steps. The user-item rating matrix is: [20 M] <table><tr><th>User / Item</th><th>I1</th><th>I2</th><th>I3</th><th>I4</th><th>I5</th></tr><tr><td>A</td><td>5</td><td>3</td><td>?</td><td>1</td><td>2</td></tr><tr><td>B</td><td>4</td><td>?</td><td>2</td><td>1</td><td>3</td></tr><tr><td>C</td><td>1</td><td>1</td><td>?</td><td>5</td><td>4</td></tr><tr><td>D</td><td>?</td><td>2</td><td>4</td><td>?</td><td>5</td></tr></table> i) Identify all common items between User A and Users B, C, and D. ii) Compute the cosine similarity: iii) Using neighbors that rated I3, calculate the predicted rating for A on I3 iv) Write the final predicted rating.	User / Item	I1	I2	I3	I4	I5	A	5	3	?	1	2	B	4	?	2	1	3	C	1	1	?	5	4	D	?	2	4	?	5	[20] [CO5]
User / Item	I1	I2	I3	I4	I5																											
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C	1	1	?	5	4																											
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